C.1.3. General probing

Table 13. MCQ

Accuracy drop	Tokens saved	Token %
14.503	1273	99.2
15.088	1238	96.4
14.620	1171	91.3
13.918	1096	85.4
10.643	1025	79.8
9.006	935	72.9
6.901	818	63.8
5.848	736	57.3
2.924	643	50.1
1.988	533	41.5
0.585	353	27.5
0.468	142	11.1
0.117	8	0.6

Table 14. Numeric-answer

Accuracy drop	Tokens saved	Token %
56.410	1940	95.6
58.974	1838	90.6
58.974	1746	86.0
55.128	1658	81.7
48.718	1572	77.4
42.308	1484	73.1
30.769	1298	63.9
28.205	1249	61.5
25.641	1146	56.4
19.231	1048	51.6
7.692	861	42.4
1.282	502	24.7
0.000	25	1.2

Table 15. Search-query

Cosine similarity	Tokens saved	Token %
0.951	197	86.2
0.958	174	76.1
0.958	161	70.4
0.964	150	65.4
0.970	137	59.8
0.979	125	54.5
0.981	112	49.0
0.984	106	46.4
0.987	97	42.3
0.991	86	37.7
0.993	72	31.3
0.999	36	15.8
1.000	1	0.6

C.2. Qwen/Qwen3-8B

Tables 16-24 show the results of early stopping experiments for Qwen/Qwen3-8B. We plot these results in Figure 5.

C.2.1. Random probing

Table 16. MCQ

Accuracy drop	Tokens saved	Token %
24.390	1764	99.5
24.429	1762	99.4
24.527	1759	99.2
24.527	1755	99.0
24.473	1751	98.8
24.571	1745	98.4
24.732	1735	97.9
24.707	1721	97.0
24.141	1690	95.3
22.459	1608	90.7
19.107	1468	82.8
9.112	854	48.2
0.488	0	0.0

Table 17. Numeric-answer

Accuracy drop	Tokens saved	Token %
59.722	2920	99.3
59.306	2918	99.3
59.028	2914	99.1
59.306	2910	99.0
59.167	2905	98.8
58.611	2898	98.6
58.333	2886	98.2
59.444	2872	97.7
58.750	2839	96.6
55.000	2734	93.0
50.556	2559	87.1
24.792	1588	54.0
0.000	0	0.0

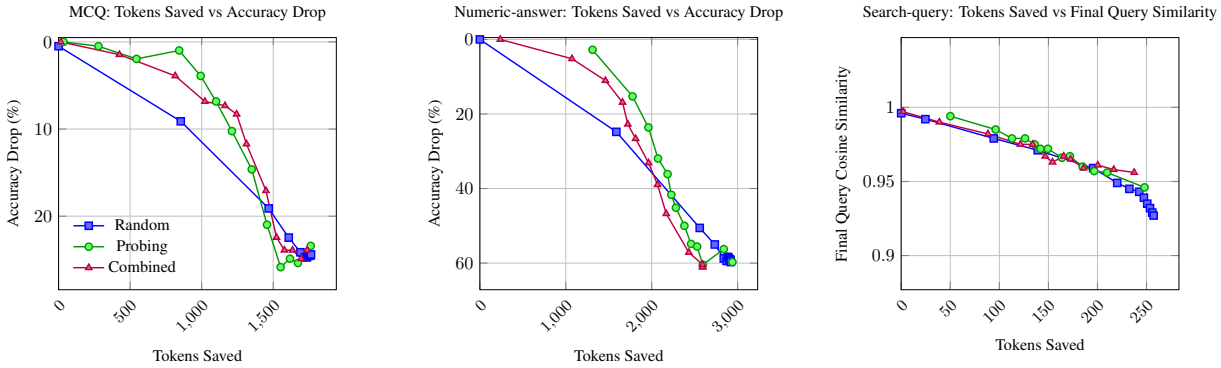


Figure 5. Early stopping performance for random, task-specific and generic probe-based gates on Qwen3-8B reasoning.

Table 18. Search-query

Cosine similarity	Tokens saved	Token %
0.927	257	92.5
0.929	256	92.0
0.932	254	91.2
0.935	251	90.2
0.939	247	89.0
0.943	242	87.2
0.945	233	83.7
0.949	220	79.2
0.959	196	70.4
0.971	139	50.0
0.979	94	33.9
0.992	25	8.9
0.996	0	0.0

Table 20. Numeric-answer

Accuracy drop	Tokens saved	Token %
59.722	2939	100.0
56.250	2839	96.6
60.417	2595	88.3
55.556	2528	86.0
54.861	2458	83.6
50.000	2380	81.0
45.139	2282	77.6
41.667	2228	75.8
36.111	2185	74.3
31.944	2073	70.5
23.611	1962	66.7
15.278	1777	60.5
2.778	1311	44.6

C.2.2. TASK-SPECIFIC PROBING

Table 19. MCQ

Accuracy drop	Tokens saved	Token %
23.415	1763	99.4
25.366	1673	94.4
24.878	1617	91.2
25.854	1552	87.5
20.976	1457	82.2
14.634	1350	76.1
10.244	1211	68.3
6.829	1100	62.1
3.902	993	56.0
0.976	843	47.5
1.951	545	30.7
0.488	279	15.7
0.000	34	1.9

Table 21. Search-query

Cosine similarity	Tokens saved	Token %
0.946	248	89.2
0.956	210	75.5
0.957	197	70.7
0.960	185	66.5
0.967	172	61.8
0.966	164	58.9
0.972	150	53.8
0.972	142	51.1
0.975	136	48.9
0.979	126	45.4
0.979	113	40.6
0.985	96	34.6
0.994	50	18.0

C.2.3. General probing

Table 22. MCQ

Accuracy drop	Tokens saved	Token %
23.902	1737	98.0
24.878	1695	95.6
23.902	1635	92.2
23.902	1578	89.0
22.439	1523	8app:aux-datasets5.9
17.073	1449	81.7
11.707	1313	74.1
8.293	1244	70.2
7.317	1163	65.6
6.829	1023	57.7
3.902	815	46.0
1.463	425	24.0
0.000	16	0.9

Table 23. Numeric-answer

Accuracy drop	Tokens saved	Token %
61.039	2593	100.0
61.039	2593	100.0
60.390	2591	99.9
57.143	2432	93.8
46.753	2169	83.7
38.961	2068	79.8
33.117	1964	75.7
26.623	1813	69.9
22.727	1722	66.4
16.883	1660	64.0
11.039	1462	56.4
5.195	1072	41.4
0.000	238	9.2

Table 24. Search-query

Cosine similarity	Tokens saved	Token %
0.956	237	85.4
0.958	217	77.9
0.961	200	72.1
0.959	186	66.8
0.965	173	62.1
0.967	166	59.6
0.963	154	55.5
0.967	147	52.8
0.975	134	48.2
0.975	121	43.7
0.982	88	31.8
0.990	39	14.0
0.997	2	0.8

D. Answer Trajectory Metrics

We plot the distributions of Answer Switches, Transient Answer Flips, and the Tokens after final switch as defined in Section 3.3

D.1. Qwen3-4B

Figures 6-17 show distribution of metrics on Qwen3-8B.

D.1.1. MCQ

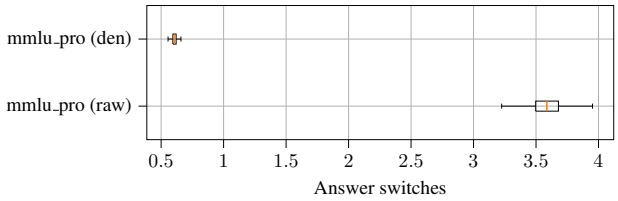


Figure 6. Distribution of the mean of answer switches for raw and denoised outputs in Qwen3-4B, evaluated on the MCQ Questions.

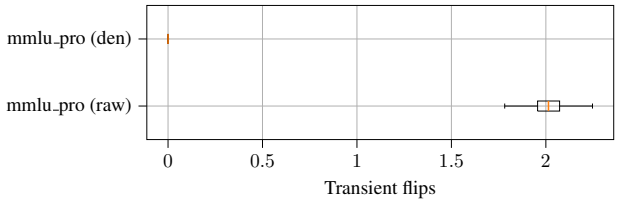


Figure 7. Distribution of the mean of transient flips for raw and denoised outputs in Qwen3-4B, evaluated on the MCQ Questions.

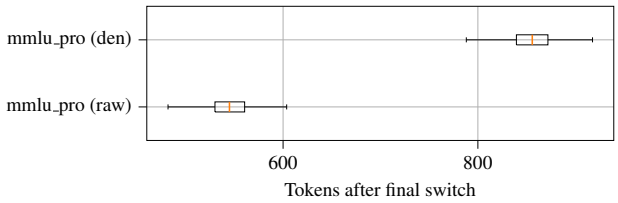


Figure 8. Distribution of the mean of tokens after final answer switch for raw and denoised outputs in Qwen3-4B, evaluated on the MCQ Questions.

D.1.2. Numeric

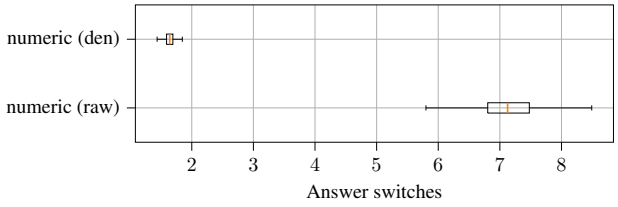


Figure 9. Distribution of the mean of answer switches for raw and denoised outputs in Qwen3-4B, evaluated on the Numeric Answers.

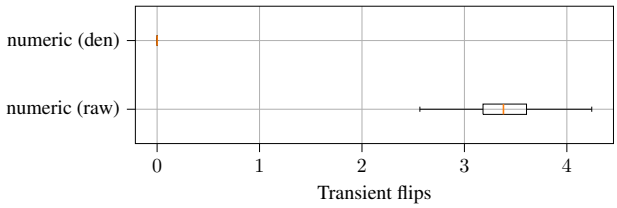


Figure 10. Distribution of the mean of transient flips for raw and denoised outputs in Qwen3-4B, evaluated on the Numeric Answers.

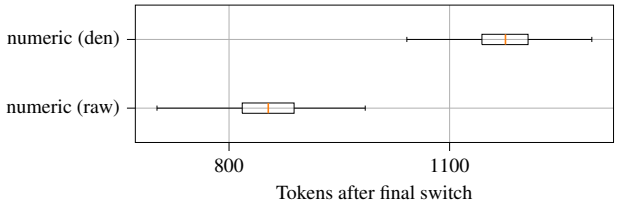


Figure 11. Distribution of the mean of tokens after final answer switch for raw and denoised outputs in Qwen3-4B, evaluated on the Numeric Answers.

D.1.3. SEARCH

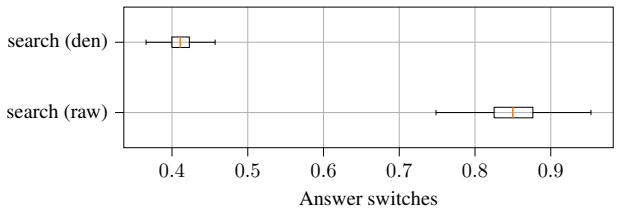


Figure 12. Distribution of the mean of answer switches for raw and denoised outputs in Qwen3-4B, evaluated on the Search Query.

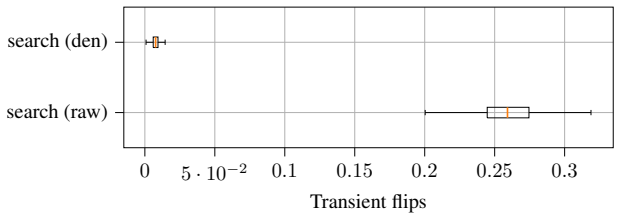


Figure 13. Distribution of the mean of transient flips for raw and denoised outputs in Qwen3-4B, evaluated on the Search Query.

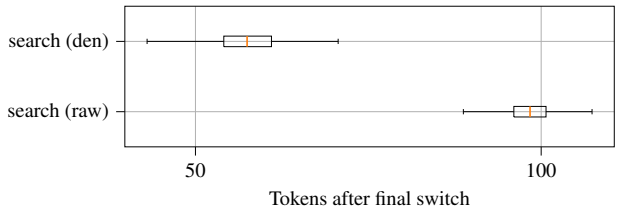


Figure 14. Distribution of the mean of tokens after final answer switch for raw and denoised outputs in Qwen3-4B, evaluated on the Search Query.

D.1.4. TOOL SELECTION

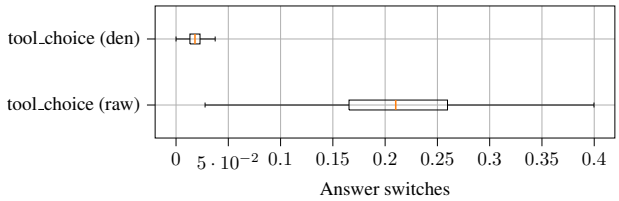


Figure 15. Distribution of the mean of answer switches for raw and denoised outputs in Qwen3-4B, evaluated on the Tool Selection.

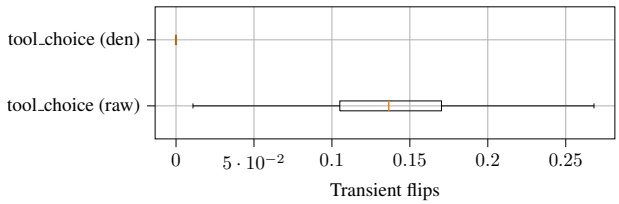


Figure 16. Distribution of the mean of transient flips for raw and denoised outputs in Qwen3-4B, evaluated on the Tool Selection.

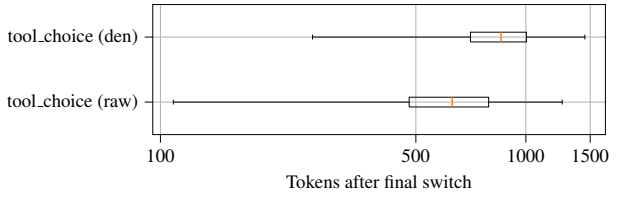


Figure 17. Distribution of the mean of tokens after final answer switch for raw and denoised outputs in Qwen3-4B, evaluated on the Tool Selection.

D.2. Qwen3-8B

Figures 18-29 shows distribution of metrics on Qwen3-8B.